GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST

Academic Period: July 2026

Program	:	**NETWORK SECURITY**
Course	:	**NETWORK MANAGEMENT AND MONITORING SYSTEMS**
Code	:	**NTC1052**
Duration	:	**3 HOURS**
Total Marks	:	**100**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

This paper consists of **6** printed pages, including this page.

Student's Name	:	**ABDUL HARIZ BIN MOHD SARNUDIN**
Student's I/D No.	:	**CBS24070208**
Semester/Trade/Group	:	**SEM 5 / DCBS / CBS 1**
Date	:	**20 / 10 / 2026**

Copyright of the German-Malaysian Institute. All rights reserved.

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Core Service Setup & Web Security	2	1		30			
2	Localhost Monitoring & Incident Automation	2	1		30			
3	Remote Agentless & Security Auditing	2	1		30			
4	System Troubleshooting & Command Line Verification	2	1		10			
				Total	100			
				Total Marks				
Total			**PO**					
			PO					
Signature								

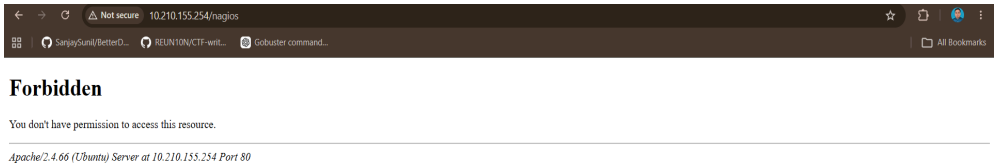
Copyright of the German-Malaysian Institute. All rights reserved.

Section 1: Core Service Setup & Web Security (30 Marks)

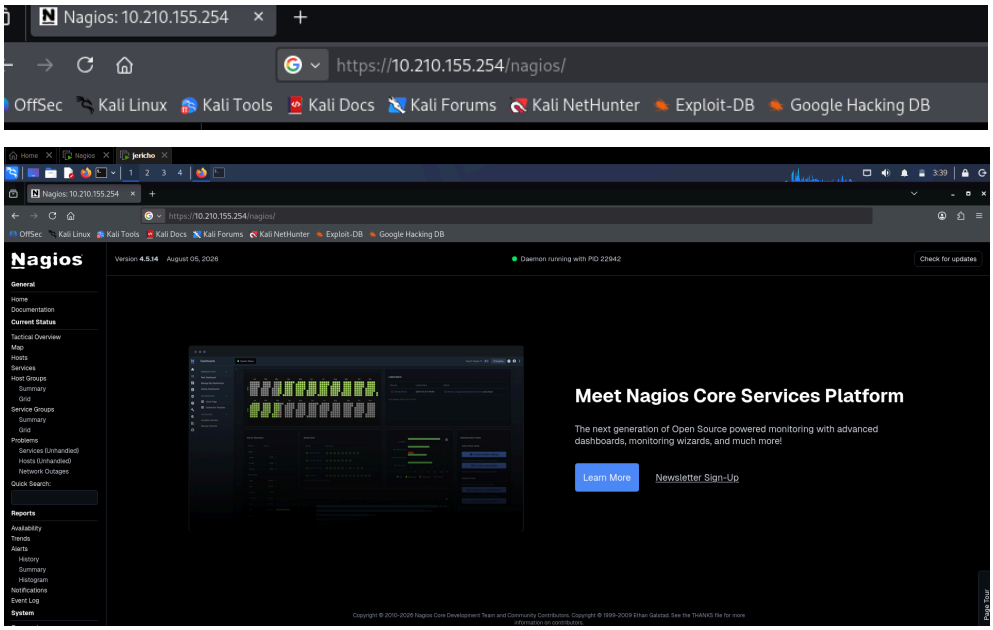
You are required to set up and secure a Nagios Core monitoring node on your designated openSUSE virtual machine.

1. Configure SSL/TLS to enforce HTTPS access for the Nagios web dashboard. Update your Apache web server configuration to restrict access to the Nagios directory, allowing connections strictly from the `192.168.1.0/24` IP range. **Capture screenshot evidence of your Apache configuration file rules and a browser window demonstrating the secure HTTPS session**. (15 Marks)

   Access externally from an external IP :



   Access from the same IP range (Client) :



Copyright of the German-Malaysian Institute. All rights reserved.

Apache2 nagios configuration file @ /etc/apache2/sites-enabled/nagios.conf.

```
        Options ExecCGI
        AllowOverride None
        <IfVersion >= 2.3>
            <RequireAll>
                Require ip 10.210.155.0/24

                AuthName "Nagios Access"
                AuthType Basic
                AuthUserFile /usr/local/nagios/etc/htpasswd.users
                Require valid-user
            </RequireAll>
        </IfVersion>
        <IfVersion < 2.3>
            Order allow,deny
            Allow from 10.210.155.0/24

            AuthName "Nagios Access"
            AuthType Basic
            AuthUserFile /usr/local/nagios/etc/htpasswd.users
            Require valid-user
        </IfVersion>
        </Directory>

        Alias /nagios "/usr/local/nagios/share"

        <Directory "/usr/local/nagios/share">
            SSLRequireSSL
            Options None
            AllowOverride None
            <IfVersion >= 2.3>
                <RequireAll>
                    Require ip 10.210.155.0/24

                    AuthName "Nagios Access"
                    AuthType Basic
                    AuthUserFile /usr/local/nagios/etc/htpasswd.users
                    Require valid-user
                </RequireAll>
            </IfVersion>
            <IfVersion < 2.3>
                Order allow,deny
                Allow from 10.210.155.0/24

                AuthName "Nagios Access"
                AuthType Basic
                AuthUserFile /usr/local/nagios/etc/htpasswd.users
                Require valid-user
            </IfVersion>
        </Directory>
        nagios@nagiosserver:/etc/apache2$ _
```


Copyright of the German-Malaysian Institute. All rights reserved.

2. Establish a log management policy for Nagios using `logrotate`. The rule set must rotate logs daily, retain a history of 7 backup cycles, and compress archived logs. **Capture screenshot evidence of your** `/etc/logrotate.d/nagios` **configuration file along with the execution output of a dry-run test** (`logrotate -d /etc/logrotate.d/nagios`). (15 Marks)

   Config file :

```
nagios@nagiosserver:~$ cat /etc/logrotate.d/nagios
/usr/local/nagios/var/nagios.log /var/log/nagios/nagios.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
    create 0664 nagios nagios
    postrotate
        /usr/bin/systemctl reload nagios > /dev/null 2>&1 || true
    endscript
}
nagios@nagiosserver:~$
```

   Dry-run test :

```
nagios@nagiosserver:~$ sudo logrotate -d /etc/logrotate.d/nagios
warning: logrotate in debug mode does nothing except printing debug messages!  Consider using verbose mode (-v) instead if this is not what you want

reading config file /etc/logrotate.d/nagios
Reading state from file: /var/lib/logrotate/status
state file /var/lib/logrotate/status does not exist
Allocating hash table for state file, size 64 entries

Handling 1 logs

rotating pattern: /usr/local/nagios/var/nagios.log /var/log/nagios/nagios.log after 1 days empty log files are not rotated, (7 rotations), old logs
considering log /usr/local/nagios/var/nagios.log
error: skipping "/usr/local/nagios/var/nagios.log" because parent directory has insecure permissions (It's world writable or writable by group which
t") Set "su" directive in config file to tell logrotate which user/group should be used for rotation.
Creating new state
considering log /var/log/nagios/nagios.log
Creating new state
nagios@nagiosserver:~$ _
```


Copyright of the German-Malaysian Institute. All rights reserved.

Section 2: Localhost Monitoring & Incident Automation (30 Marks)

Configure active service checks and automated self-healing mechanisms for the local monitoring system (localhost).

1. Define checks for local HTTP response time (Warning: >200ms, Critical: >500ms) and swap space utilization (Warning: >50%, Critical: >80%). Restrict alert notification windows to standard business hours (Monday to Friday, 09:00 to 17:00). **Capture screenshot evidence showing these active service definitions within the Nagios web console.** (15 Marks)

   Business hours config :

```
nagios@nagiosserver:~$ cat /usr/local/nagios/etc/objects/timeperiods.cfg | tail
define timeperiod {

    timeperiod_name    businesshours
    alias              Standard Business Hours
    monday             09:00-17:00
    tuesday            09:00-17:00
    wednesday          09:00-17:00
    thursday           09:00-17:00
    friday             09:00-17:00
}
nagios@nagiosserver:~$
```

   HTTP response time and swap space utilization :

```
nagios@nagiosserver:~$ cat /usr/local/nagios/etc/objects/localhost.cfg | tail -20
#HTTP response time: Warning >200ms, Critical>500ms
#Swap Space utilization: Warning >50% used and Critical >80% used

define service {
    use                     local-service
    host_name               localhost
    service_description     HTTP Response Time
    check_command           check_http!-w 0.200 -c 0.500
    notification_period     businesshours
    event_handler           restart_apache
    event_handler_enabled   1
}

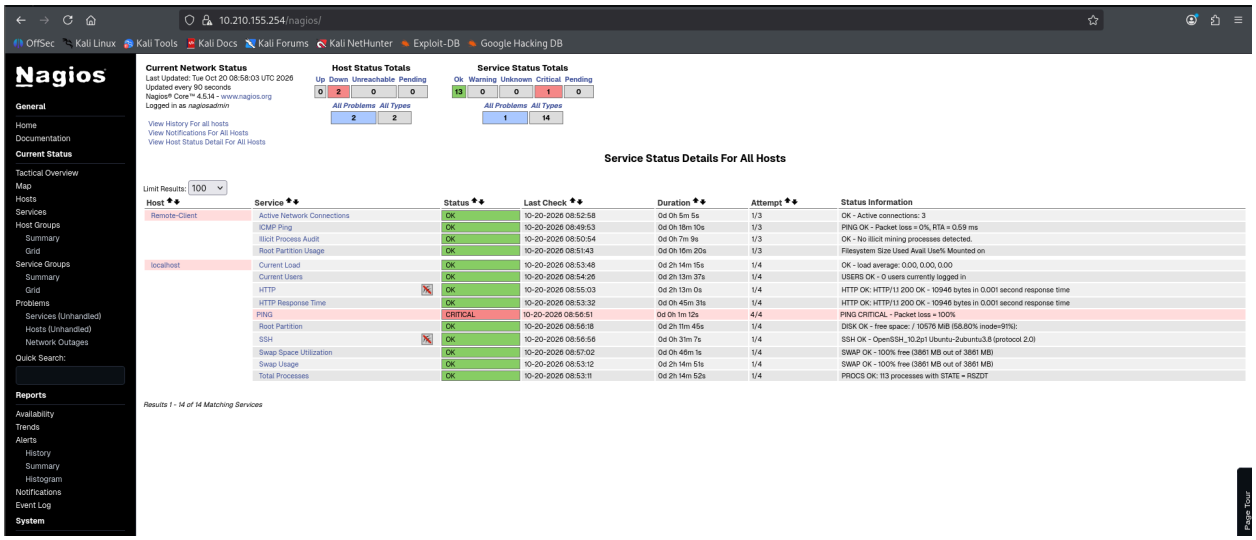
define service {
    use                     local-service
    host_name               localhost
    service_description     Swap Space Utilization
    check_command           check_swap!50%!20%
    notification_period     businesshours
}
nagios@nagiosserver:~$ _
```


Copyright of the German-Malaysian Institute. All rights reserved.

Nagios web console :

localhost	Current Load	OK	10-20-2026 08:53:48	0d 2h 14m 15s	1/4	OK - load average: 0.00, 0.00, 0.00
	Current Users	OK	10-20-2026 08:54:26	0d 2h 13m 37s	1/4	USERS OK - 0 users currently logged in
	HTTP	OK	10-20-2026 08:55:03	0d 2h 13m 0s	1/4	HTTP OK: HTTP/1.1 200 OK - 10946 bytes in 0.001 second response time
	HTTP Response Time	OK	10-20-2026 08:53:32	0d 0h 45m 31s	1/4	HTTP OK: HTTP/1.1 200 OK - 10946 bytes in 0.001 second response time
	PING	CRITICAL	10-20-2026 08:56:51	0d 0h 1m 12s	4/4	PING CRITICAL - Packet loss = 100%
	Root Partition	OK	10-20-2026 08:56:18	0d 2h 11m 45s	1/4	DISK OK - free space: / 10576 MiB (58.80% inode=91%):
	SSH	OK	10-20-2026 08:56:56	0d 0h 31m 7s	1/4	SSH OK - OpenSSH_10.2p1 Ubuntu-2ubuntu3.8 (protocol 2.0)
	Swap Space Utilization	OK	10-20-2026 08:57:02	0d 0h 46m 1s	1/4	SWAP OK - 100% free (3861 MB out of 3861 MB)
	Swap Usage	OK	10-20-2026 08:53:12	0d 2h 14m 51s	1/4	SWAP OK - 100% free (3861 MB out of 3861 MB)
	Total Processes	OK	10-20-2026 08:53:11	0d 2h 14m 52s	1/4	PROCS OK: 113 processes with STATE = RSZDT

Results 1 - 14 of 14 Matching Services



2. Construct a custom Bash script to act as an event handler. When the HTTP service transitions into a `CRITICAL` state, the script must automatically trigger a service restart (`systemctl restart apache2`). **To verify implementation, manually stop the HTTP service, allow Nagios to detect the failure, and capture screenshot evidence from the Nagios Event Log showing the automated restart action.** (15 Marks)

Event Logs signifying apache2 stopped and restarted :

Copyright of the German-Malaysian Institute. All rights reserved.

Current Event Log
Last Updated: Tue Oct 20 08:14:31 UTC 2026
Nagios® Core™ 4.5.14 - www.nagios.org
Logged in as *nagiosadmin*

Latest Archive

Log File Navigation
Tue Oct 20 00:00:00 UTC 2026
to
Present..

File: /usr/local/nagios/var/nagios.log

October 20, 2026 08:00

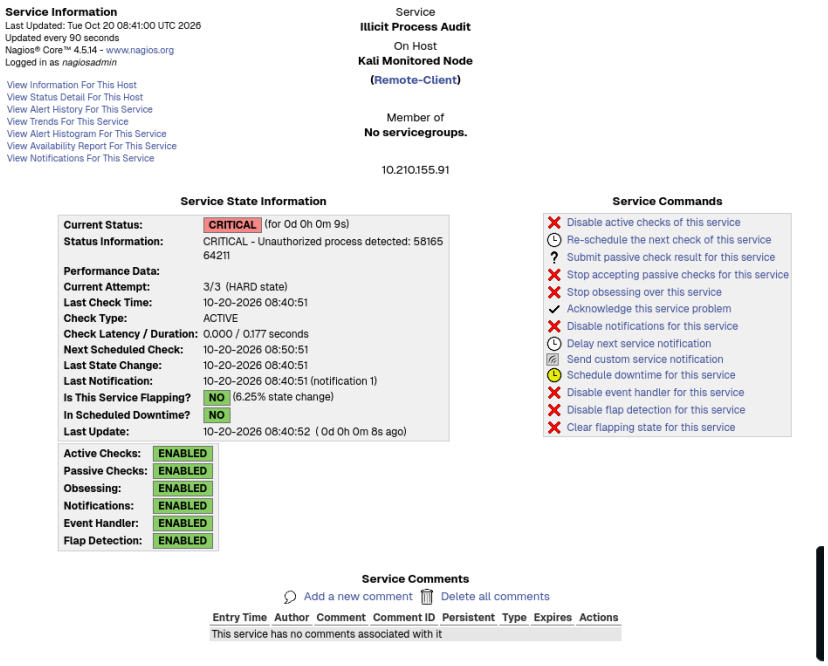
S [10-20-2026 08:12:32] SERVICE EVENT HANDLER: localhost;HTTP Response Time;OK;SOFT;2;restart_apache
[10-20-2026 08:12:32] SERVICE ALERT: localhost;HTTP Response Time;OK;SOFT;2;HTTP OK: HTTP/1.1 200 OK - 10946 bytes in 0.001 second response time
S [10-20-2026 08:11:32] SERVICE EVENT HANDLER: localhost;HTTP Response Time;CRITICAL;SOFT;1;restart_apache
! [10-20-2026 08:11:32] SERVICE ALERT: localhost;HTTP Response Time;CRITICAL;SOFT;1;connect to address 127.0.0.1 and port 80: Connection refused

Copyright of the German-Malaysian Institute. All rights reserved.

Section 3: Remote Agentless & Security Auditing (30 Marks)

Configure Nagios to monitor a secondary host on the network and implement custom security monitoring routines.

1. Configure basic connectivity checks for the neighboring host using ICMP ping (Warning: >5% packet loss, Critical: >10% packet loss) and set a critical alert threshold for root partition (/) disk usage when free space falls below 10%. **Capture screenshot evidence of the Nagios web interface showing the current status of the remote host.** (15 Marks)

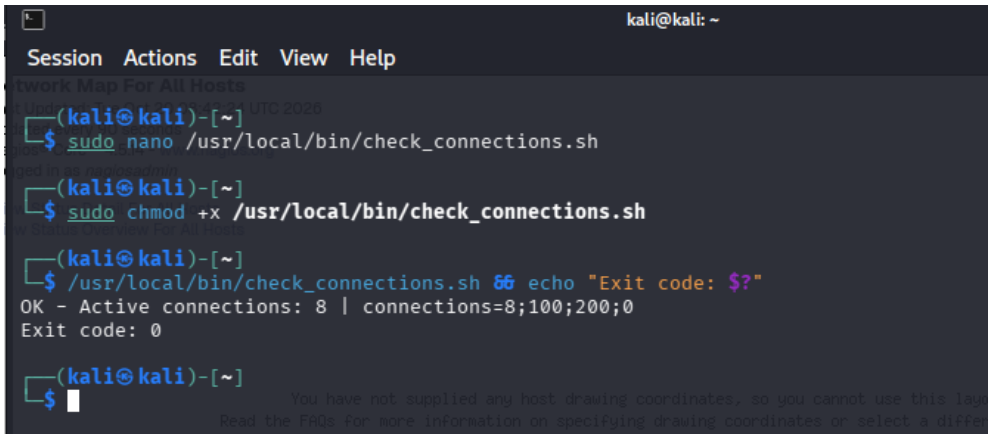


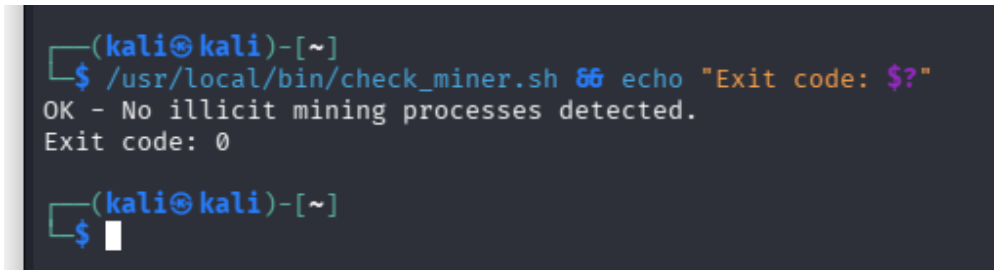


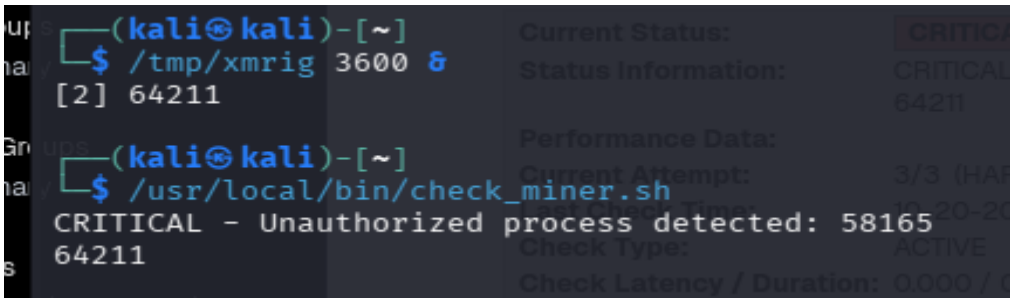
Copyright of the German-Malaysian Institute. All rights reserved.

2. Write two custom Nagios plugin scripts and deploy them to monitor the remote host:
    - A script counting active network connections, setting Warning at >100 connections and Critical at >200 connections.
    - A script checking for unauthorized process names (e.g., `xmrig` or `minerd`), returning a `CRITICAL` state immediately if detected.

Simulate an unauthorized process execution on the remote system and capture screenshot evidence of the triggered alert on the Nagios dashboard. (15 Marks)







Copyright of the German-Malaysian Institute. All rights reserved.

Scripts :

```
┌──(kali㉿kali)-[~]
└─$ cat /usr/local/bin/check_connections.sh /usr/local/bin/check_miner.sh
#!/bin/bash
# Nagios Plugin: Active Network Connections
# Thresholds: Warning > 100, Critical > 200

CONNS=$(ss -tun | grep -E "ESTAB|SYN-SENT|SYN-RECV" | wc -l)

if [ "$CONNS" -gt 200 ]; then
    echo "CRITICAL - Active connections: $CONNS | connections=$CONNS;100;200;0"
    exit 2
elif [ "$CONNS" -gt 100 ]; then
    echo "WARNING - Active connections: $CONNS | connections=$CONNS;100;200;0"
    exit 1
else
    echo "OK - Active connections: $CONNS | connections=$CONNS;100;200;0"
    exit 0
fi
#!/bin/bash
# Nagios Plugin: Illicit Process Check (xmrig/minerd)

FOUND=$(pgrep -f "xmrig|minerd")

if [ -n "$FOUND" ]; then
    PROCS=$(ps -fp $FOUND -o pid,comm= | tr '\n' ' ')
    echo "CRITICAL - Unauthorized process detected: $PROCS"
    exit 2
else
    echo "OK - No illicit mining processes detected."
    exit 0
fi

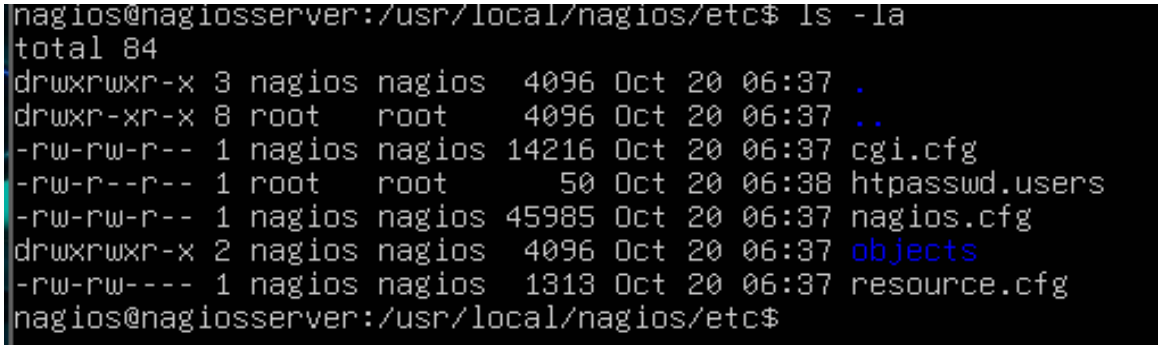
┌──(kali㉿kali)-[~]
└─$
```


Copyright of the German-Malaysian Institute. All rights reserved.

Section 4: System Troubleshooting & Command Line Verification (10 Marks)

Complete the following operational tasks directly on your server and provide the corresponding verification evidence.

1. Using the command line, locate the main `nagios.cfg` file on your openSUSE system. Display its file path, access permissions, owner, and modification date. Provide a screenshot of the terminal output. (1 Mark)



2. Write out a complete, syntax-valid Nagios configuration block defining a new host object (`Core-Switch`, IP: `192.168.1.254`) and its corresponding ICMP ping service object. (1 Mark)

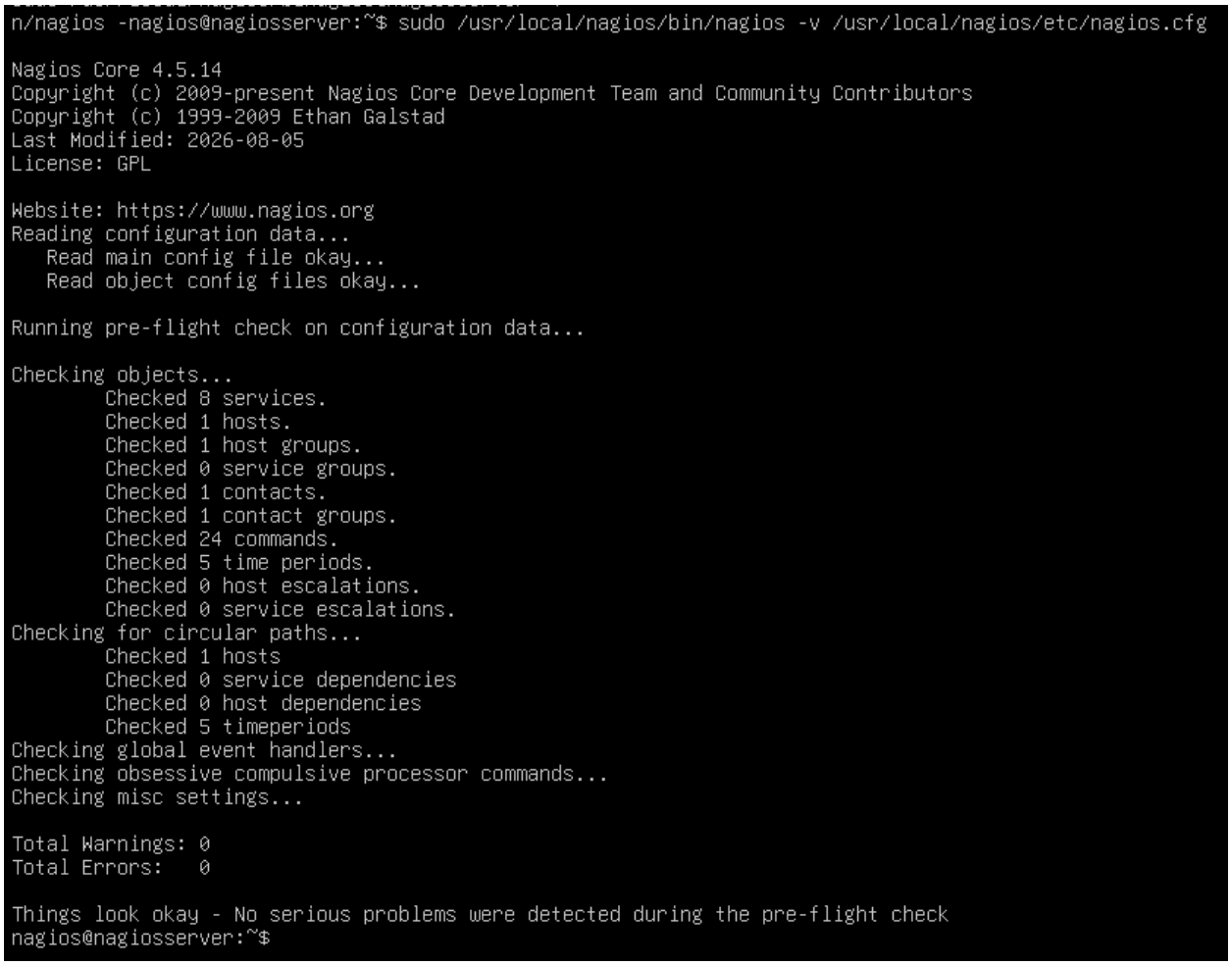
```
define host {
   use                 generic-switch
   host_name           Core-Switch
   alias               Core Backbone Infrastructure Switch
   address             192.168.1.254
   max_check_attempts  5
   check_period        24x7
   notification_period 24x7
   notification_interval 30
}
```


Copyright of the German-Malaysian Institute. All rights reserved.

```
define service {
    use                     generic-service
    host_name               Core-Switch
    service_description     ICMP Ping
    check_command           check_ping!100.0,20%!500.0,60%
}
```

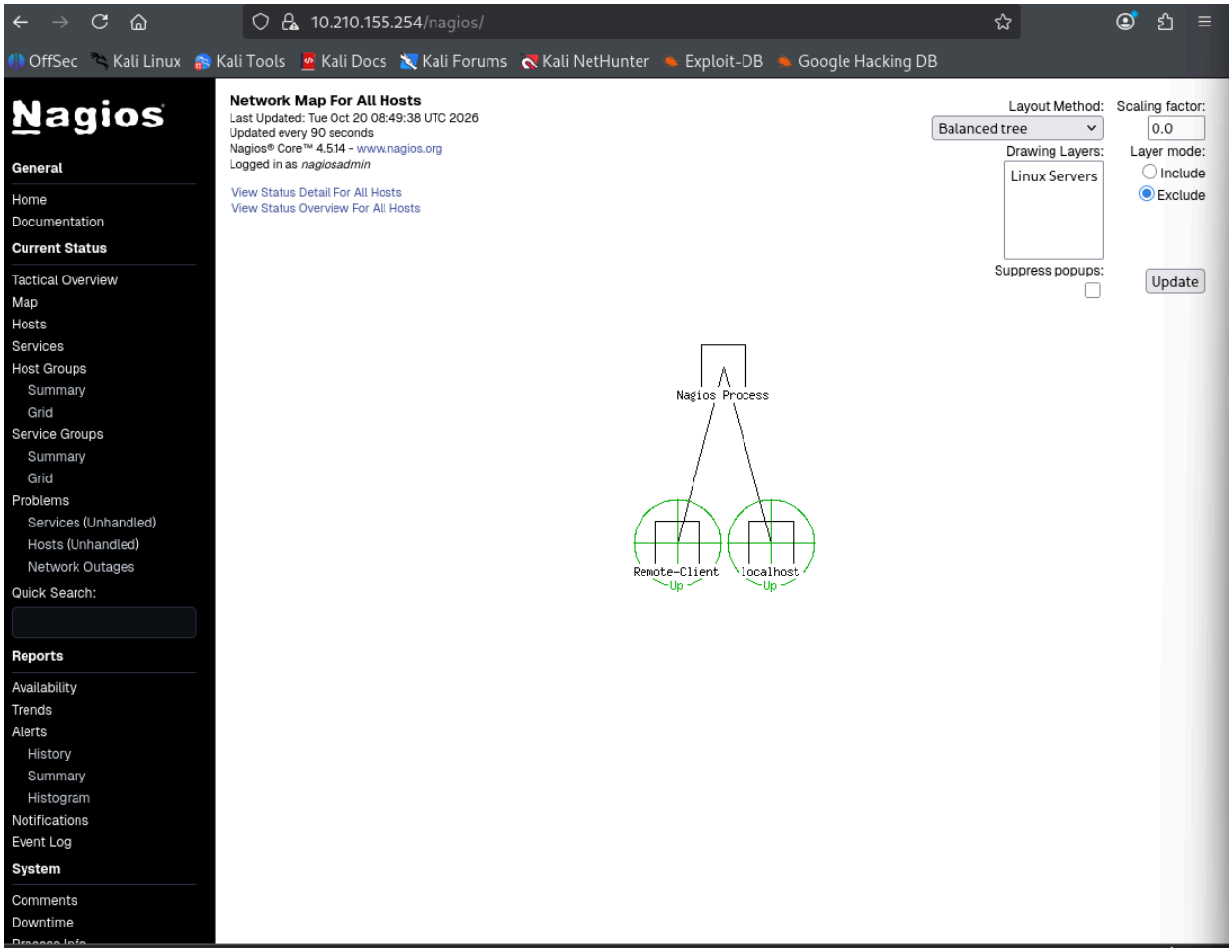

Copyright of the German-Malaysian Institute. All rights reserved.

3. Run the Nagios configuration verification command (`nagios -v /etc/nagios/nagios.cfg`) via the terminal. Provide a screenshot showing zero warnings and zero errors. (1 Mark)



Copyright of the German-Malaysian Institute. All rights reserved.

4. Navigate to the Status Map on the Nagios web interface. Capture a screenshot, explain how the map represents host states, and briefly describe what the visual status indicators signify. (1 Mark)



The Status Map presents a hierarchical network topology graph radiating from the central Nagios monitoring process, illustrating parent-child dependencies and reachability relationships among network nodes.

Copyright of the German-Malaysian Institute. All rights reserved.

5. Execute one of your custom plugin scripts directly from the terminal (e.g.,`/check_disk -w 20% -c 10% -p /`). Immediately run `echo $?` To display the exit code (0 for OK, 1 for Warning, 2 for Critical). Provide a screenshot of the terminal output. (1 Mark)

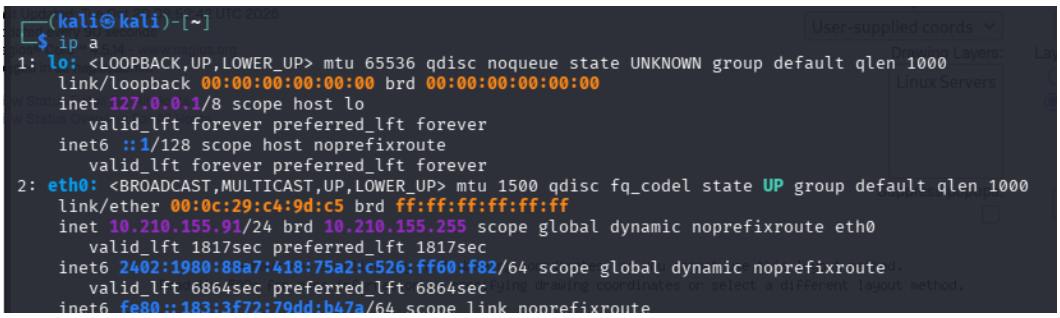
```
nagios@nagiosserver:~$ /usr/local/nagios/libexec/check_disk -w 20% -c 10% -p /

echo $?DISK OK - free space: / 10576 MiB (58.80% inode=91%);| /=7409MiB;15179;17076;0;18974
nagios@nagiosserver:~$
```

6. Perform an active incident analysis on the neighboring host: (5 Marks)

1. Block ICMP network traffic on the target machine or bring down its network interface.

   Client IP :



   Ping from Server failed 100% packet lost :

```
nagios@nagiosserver:~$ ping -c 2 10.210.155.91
PING 10.210.155.91 (10.210.155.91) 56(84) bytes of data.

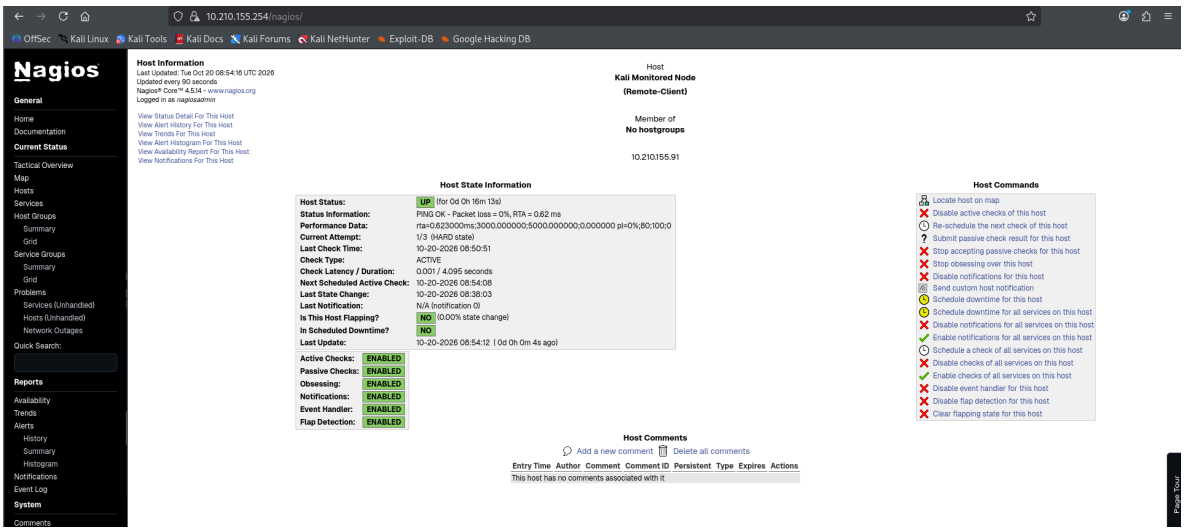
--- 10.210.155.91 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1033ms

nagios@nagiosserver:~$ _
```

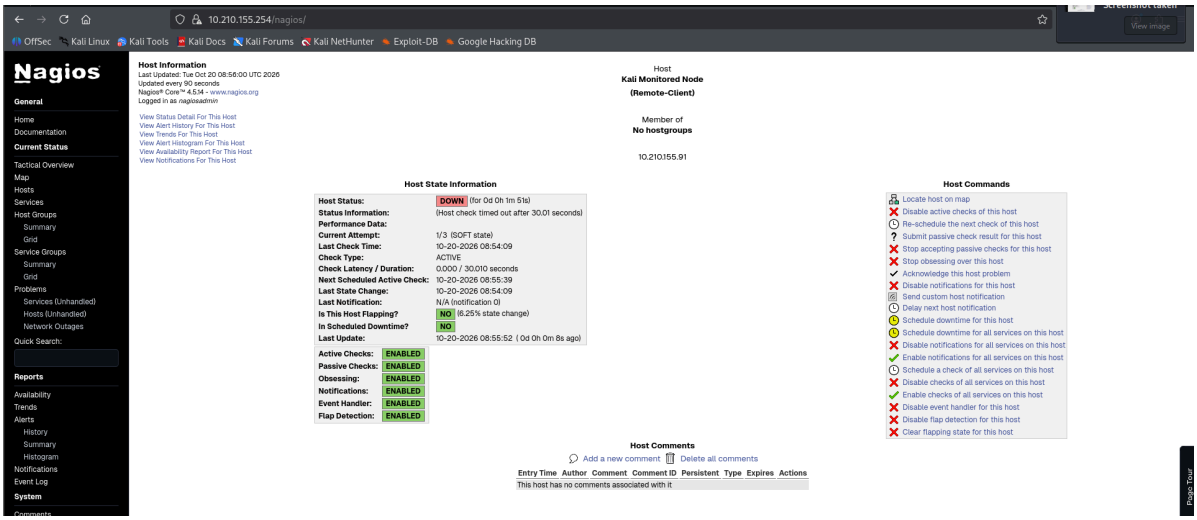

Copyright of the German-Malaysian Institute. All rights reserved.

2. Capture a screenshot of the Nagios Host Detail page showing the host state transition from `UP` to `DOWN`.

   Before recheck of the host :



   After recheck of the host :



3. Display the last 10 lines of `/var/log/nagios/nagios.log` in your terminal, documenting the exact time of the detected failure. Provide screenshot evidence.

```
nagios@nagiosserver:~$ ^C
nagios@nagiosserver:~$ tail -n 10 /usr/local/nagios/var/nagios.log
[1792486378] wproc:   stderr line 02: printf: write error: Broken pipe (os error 32)
[1792486449] EXTERNAL COMMAND: SCHEDULE_FORCED_HOST_CHECK;Remote-Client;1792486448
[1792486479] wproc: Core Worker 27679: job 14 (pid=28043) timed out. Killing it
[1792486479] wproc: CHECK job 14 from worker Core Worker 27679 timed out after 30.01s
[1792486479] wproc:   host=Remote-Client; service=(null);
[1792486479] wproc:   early_timeout=1; exited_ok=0; wait_status=0; error_code=62;
[1792486479] wproc:   stdout line 01: PING CRITICAL - Packet loss = 100%|rta=5000.000000ms;3000.000000;5000.000000;0.000000 pl=100%;80;100;0
[1792486479] Warning: Check of host 'Remote-Client' timed out after 30.01 seconds
[1792486479] HOST ALERT: Remote-Client;DOWN;SOFT;1;(Host check timed out after 30.01 seconds)
[1792486479] wproc: Core Worker 27679: job 14 (pid=28043): Dormant child reaped
nagios@nagiosserver:~$
```


Copyright of the German-Malaysian Institute. All rights reserved.